Fiscal Year:	FY 2011	**Task Last Updated:**	FY 06/08/2011
PI Name:	Alfano, Robert Ph.D.		
Project Title:	Biophotonic Plant (Moss) Stress Detection		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	ralfano@ccny.cuny.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	212-650-5531
Organization Name:	City College of City University of New York		
PI Address 1:	160 Convent Ave		
PI Address 2:	MR 419		
PI Web Page:			
City:	New York	**State:**	NY
Zip Code:	10031-9101	**Congressional District:**	15
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Space Biology--Microbial, Plant, Cell Biology: NNH08ZTT003N
Start Date:	06/09/2009	**End Date:**	12/08/2010
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/08/2010 (previously 6/08/2010) per NSSC (08/2010)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX09AL97G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The objective of this research is to study the effect of gravity on the cytoplasmic and chloroplastic calcium profile in aequorin-transfomed Moss, Physcomitrella patens, using bioluminescence measurement with genetic reporter. This multi-disciplinary research program will focus on understanding the effect of gravity on calcium release in cytoplasm and chloroplast using photonic spectroscopy technology and transgenic techniques. The main effects to be studied are 1) the role of gravistimulation on the change in calcium release by in situ measurement of bioluminescence from moss expressing apoaequorin mounted on a traditional clinostat, 2) investigate, in situ, the change in the dynamics of cytoplasmic and chloroplastic calcium concentration in different gravitational fields by the use of centrifuges, and 3) study calcium expression under simulated zero gravity by in situ instantaneous measurements on plants in free fall. This project will set the stage for further understanding the precise mechanisms by which various unique gravity stress stimuli generate distinct Ca2+ responses that result in distinct downstream physiologic responses.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	A pilot study was conducted under the NASA grant NNX09AL97G (RF No. 49673-00-01) to research the response of certain moss strains (Physcomitrella patens) genetically modified with aequorin. Several moss samples were investigated under various stress conditions, such as, temperature variation, touch, pH factor to study how calcium Ca2+ released in cytoplasm and chloroplast reacts with aequorin and results in emission of blue bioluminescence at 470nm within seconds of the onset of stress. Our repeated efforts with different moss strains created from the same batch of aequorin transformation using several detection systems failed to yield positive response.
Bibliography Type:	Description: (Last Updated:)